

Technology Stack

Date	05 JULY 2026
Team ID	SWTID-2026-2423
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	HTML5, CSS3, JavaScript	<i>Provides a highly responsive user interface compatible across standard web browsers (Chrome/Edge) to display text inputs and tutor dashboards seamlessly.</i>

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	<i>Python 3.8+ (Flask Framework)</i>	<i>Flask is a lightweight micro-framework chosen for fast processing, easy integration with deep learning scripts, and quick local web testing via VS Code or PyCharm.</i>
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	<i>SQLite / JSON File Storage</i>	<i>Perfect for handling locally tracked text strings, query metadata, and emotion tags without requiring a heavy external database setup during development.</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>Local Host Deployment (Anaconda / Jupyter Environment)</i>	<i>Built and deployed on a local machine environment running Anaconda Navigator, utilizing an internet connection for downloading necessary deep learning datasets and packages.</i>
5	Version Control & CI/CD	<i>GitHub</i>	<i>Selected for seamless project source code backups, version tracking of model training scripts, and effortless collaboration.</i>

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Hugging Face Transformers API / PyTorch</i>	<i>Enables deep learning-driven Natural Language Processing (NLP) to directly analyze, tag, and extract emotion metrics from student queries in real time.</i>